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Recurrence Computational Fluid Dynamics (rCFD)



A Data-Driven Reduced-Order Acceleration Technique

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What is rCFD?



The Big Picture

Definition

Data-driven, reduced-order acceleration technique for simulating transport phenomena over extensive timescales.

Instead of solving Navier-Stokes at every time step, we **reconstruct** the flow field from previously calculated sequences.

- Exploits **recurrent patterns** in statistically stationary flows.
- Speed-up: **100x to 10,000x** compared to full CFD.
- Enables simulations that would otherwise take weeks → hours.

Working Mechanism

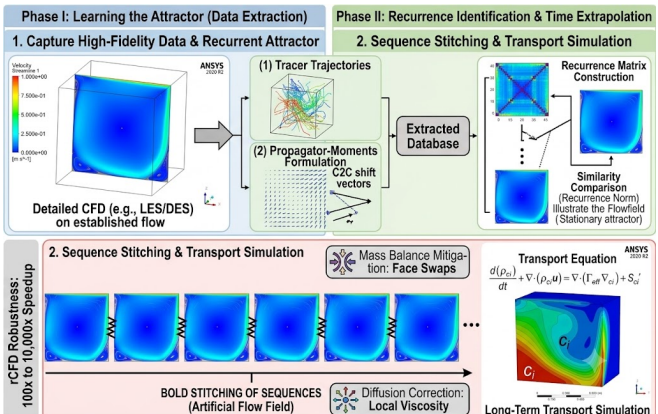


Phase I – Learning the Attractor

- Run a high-fidelity CFD simulation (LES, DES) to capture **fast dynamics**.
- Extract convection and diffusion data:
  - **Massless tracer particles** (track trajectories)
  - **Propagator-moments formulation** (passive scalar fields → cell-to-cell shift vectors)

Goal: Sample the “stationary attractor” – the set of recurring flow patterns.

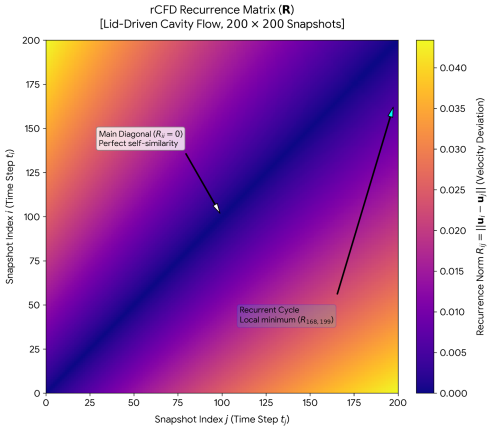
Phase II – The rCFD Acceleration Engine



Core Workflow:

1. **Extract Data:** Run high-fidelity simulation to sample attractor.
2. **Map Attractor:** Compute Recurrence Matrix ($\mathbf{R}$).
3. **Stitch Sequence:** Use $\mathbf{R}$ to find smooth flow transitions.
4. **Transport Solve:** Decouple fluid $\mathbf{u}$ from passive scalars c_i.

Phase II – Recurrence Matrix Anatomy



1. Mathematical Formulation

Each cell (i, j) represents the difference (recurrence norm) between velocity fields of two distinct time frames, t_i and t_j:

$$R_{ij} = \|\mathbf{u}_i - \mathbf{u}_j\|$$

where $\mathbf{u}$ is the velocity vector field and $\|\cdot\|$ denotes the L_2 norm.

Phase II – Recurrence Matrix Anatomy

2. Key Visual & Numerical Characteristics

- **Main Diagonal:** $R_{ii} = 0$ represents perfect self-similarity.
- **Symmetry:** $R_{ij} = R_{ji}$ (isotropic distance metric).
- **Attractor Center:** Snapshot 100 minimizes average distance (0.0108).
- **Off-Diagonal Cycles:** Minimum $R_{168,199} \approx 0.0051$ proves cycle closure.

3. Utility in rCFD

- **A. Identification:** Dense low-R valleys mark when the attractor is fully mapped.
- **B. Stitching Guide:** Links 199 → 168 for smooth, mass-conserving transitions.
- **C. Scale Isolation:** Locks velocity $\mathbf{u}$, shifting resources to transport:

Phase III – Stitching & Transport

1. **Stitch** sequences to compose an artificial, long-term flow field.
2. **Solve transport** on top of this reconstructed field:

$$\frac{\partial(\rho c_i)}{\partial t} + \nabla \cdot (\rho c_i \mathbf{u}) = \nabla \cdot (\Gamma_{\text{eff}} \nabla c_i) + S_{c_i}$$

c_i: species concentration
Γ_{eff}: effective diffusivity (molecular + turbulent)

Result: Fast fluid motion separated from slowly evolving species/thermal properties.

Robust Sides



Why rCFD is Robust

- **Computational efficiency:** Speed-up factors of **100–10,000x**.
  Example: Steel tundish simulation at 1/2700 of DES runtime – faster than real-time.
- **Data-assisted symmetrization:** Even without strong recurrences, rCFD reproduces correct average flow properties via statistical sampling.
- **Mathematical grounding:** Propagator theory provides a rigorous framework for accurate mapping, handling large velocity ranges and varying cell sizes.

Statistical fidelity is maintained; strict dynamical consistency is sacrificed for speed.

Industrial Benefits



Real-World Impact

Digital Twins & Real-Time Control

- Faster than real-time → immediate process monitoring.
- Synchronized control based on sub-plant heterogeneities.

Soft Sensors

Synchronized rCFD delivers peak temperatures, concentration gradients, etc., to control systems via virtual sensors.

Process Optimization

- Many-parameter sweeps
- Residence time distribution analyses

Required Skills



What You Need to Implement rCFD

- **Traditional CFD expertise:** Navier-Stokes solvers (SIMPLE), turbulence modelling (RANS, LES, DES).
- **Computational fluid programming:** UDFs, scripting (C, Scheme) for database automation.
- **Statistical analysis:** Convergence of leading statistical moments, recurrence prominence.
- **Validation capabilities:** Compare rCFD outputs against PIV experiments or detailed CFD predictions.

A multidisciplinary skill set – but the payoff is huge.

Limitations & Corrections



When rCFD Fails (Bold Failures)

Cannot safely extrapolate

- Strong transients / start-up phases
- Chaotic turbulence without recurring patterns
- Changing geometry or fluid rheology

→ Not a general-purpose CFD replacement

How to correct “bold failures”

- **Mass imbalance:** Mass-acting face swaps based on a global error monitor.
- **Numerical diffusion:** Account for local sub-grid viscosity and physical dispersion.
- **Low recurrence:** Domain

Summary



Take-Home Messages

rCFD is a **game changer** for long-timescale transport simulations in recurrent flows.

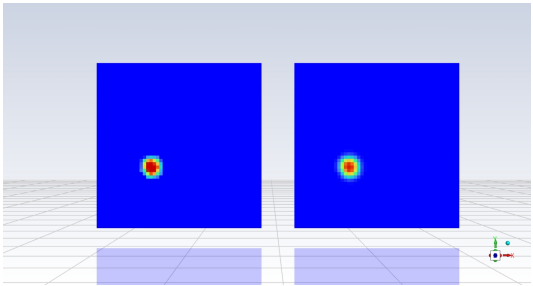
Speed	Fidelity	Use case
100–10,000x faster	Statistical (not pointwise)	Digital twins, optimisation

- Works best when flow is statistically stationary and repetitive.
- Requires physics-based corrections (mass balance, diffusion, domain decomposition).

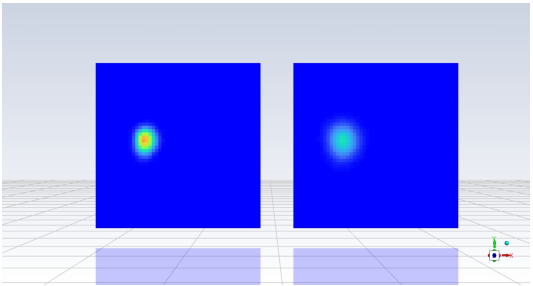
Example Predictions



Example Predictions using rCFD in Lid driven cavity



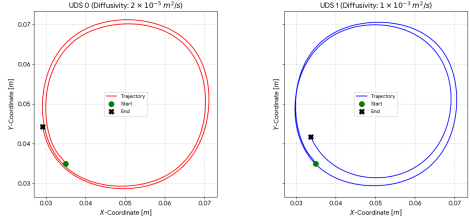
Initial State (Snapshot 0001)



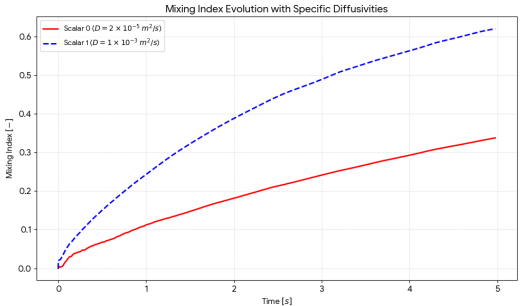
Later Mixing State (Snapshot 0013)

Visualization of scalar species mixing within the lid-driven cavity, transitioning from a localized initial condition to a more dispersed state.

Example Predictions using rCFD in Lid driven cavity



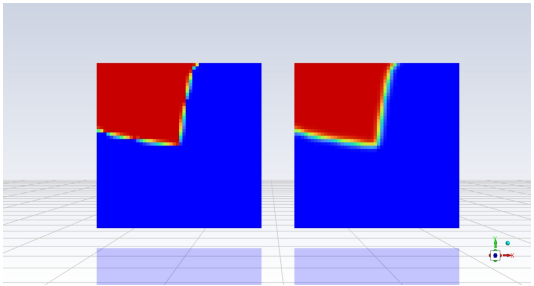
Center of Gravity (Cog) Evolution



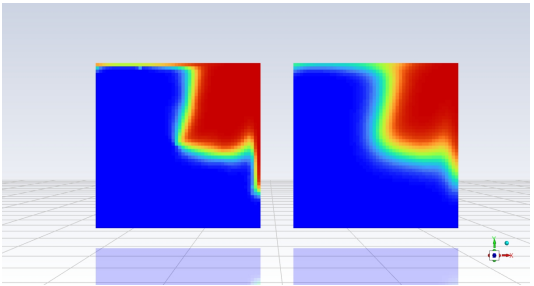
Mixing Index Evolution

Left: Tracking the global Center of Gravity of the species. Right: The mixing index approaching its asymptotic equilibrium value, demonstrating the ability of rCFD to capture long-term mixing dynamics.

Arbitrary Shape Placement & Fast Prediction



Initial Shape Definition (Snapshot 0001)



Predicted Diffusion State (Snapshot 0026)

- **Any shape, any position:** Freely define arbitrary species concentration profiles anywhere in the flow domain without re-running the fluid solver.



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